

Akash Dubey

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EDUCATION

Rutgers University Honors College, School of Arts and Sciences

New Brunswick, NJ

Bachelor of Science in Computer Science and Mathematics

Anticipated May 2027

- **GPA:** 3.95/4.0 | SAT: 1580
- **Relevant Coursework:** Tensor Networks, Systems Programming, Stochastic Processes, Computer Architecture, Algorithms, Data Structures, Honors Calculus III & IV, Math Theory of Probability.
- **Academic Programs & Research:** Directed Research Reading Program on Symmetric Functions (representation theory for combinatorial optimization); Physics Learning Theory Seminar (statistical mechanics in high-dimensional learning), Dean's List.
- **Activities:** Quantitative Finance Club, Road to Silicon Valley Program (Cohort 6), ESL Instructor, Pickleball.

RESEARCH

Algorithmic Robotics and Control Lab in Rutgers's Dept. of CS

New Brunswick, NJ

Research Assistant (Advisor: Prof. Jingjin Yu)

Jan 2026 – Present

- Implemented and validated CUDA versions of pRRTC and FCIT*, parallelizing tree/graph search and collision checking for high-dimensional ASAO robot motion planning across 500 benchmark problems.
- Developing a graph-based search policy (GaP) for dual-arm task-and-motion planning (TAMP), learning from successful and failed planner traces to rank symbolic action skeletons and predict geometric feasibility. Working in Isaac Sim specifically in dual-arm motion planning to further bridge TAMP capabilities with ML.

Kwan Lab, Rutgers Cancer Institute of New Jersey

New Brunswick, NJ

Computational Biology Research Assistant (Advisor: Prof. Kelvin Kwan)

Jan 2026 – Present

- Configured and validated a reproducible DeepSTARR inference environment, resolving legacy TensorFlow/Keras dependencies on HPC to enable sequence-based prediction of developmental and housekeeping enhancer activity.
- Developed an AlphaGenome-based in silico deletion screen spanning 12 candidate sites, 3 deletion sizes (10, 40, and 80 kb), and 2 cell types (72 perturbation conditions) to quantify TAD-boundary sensitivity; identified a chr12 insulator with the strongest predicted insulation loss among 12 candidates.
- Built a 3D chromatin polymer simulator using overdamped Langevin dynamics to generate conformations from Hi-C contact matrices; implemented multi-scale TAD detection using insulation score and directionality index methods and validated the pipeline with 70+ automated tests.

Rutgers Economics Labs

New Brunswick, NJ

President (formerly Research Director, Team Lead)

Oct 2024 – Present

- Architected an AI automated economic research pipeline RAIL (Rutgers Agentic Intelligence Labs) to collect, organize, analyze, and synthesize economic data to derive insights using data ontologies and coding agent workflows.
- Leading quantitative research projects for partners including NJDOL, NJDEP, NJBPU, Federal HUD, U.S. Census Bureau; applied econometric models in Python/R to analyze large datasets and project economic trends.
- Previously led a team of five researchers to evaluate the efficacy of the Urban Enterprise Zone program for the NJDCA, utilizing econometric methods in R and Python to analyze complex census datasets.

SELECTED TECHNICAL PROJECTS

SLM Optimization & Post-Training (GSM8k) | *PyTorch, HPC, Quantization, Pruning*

- Fine-tuned small language models (SLMs) on the GSM8k math reasoning dataset to study the effects of different model sizes and architectures on math reasoning capabilities.
- Deployed models on the Rutgers High Performance Computing (Amarel) cluster, utilizing quantization and structural pruning to maximize inference throughput on constrained compute.

Knowledge Base and AI Agent Workflow Manager | *Data Ontology, Workflows, Coding Agents*

- Built KRAIL (pypi), a local-first, repo-backed knowledge runtime that structures raw notes into ontology-aware topic pages, markdown graphs, source ledgers, claim records, verification runs, and artifact lineage for evidence-grounded agent memory.

- Implemented MCP/CLI and agent infrastructure for auditable coding-agent workflows (for software factories, auto-research, and company brain use cases), including local runner dispatch, dry-run work orders, session logs, role prompts/checklists, workflow validation, and hybrid keyword-graph-vector retrieval.

Local Coworker: Native macOS Computer-Use Agent | *Swift, MCP, Gemini Vision* (private code)

- Built a native desktop agent capable of autonomous computer use by integrating Gemini Vision with macOS Accessibility APIs, AppleScript, and JXA.
- Implemented the Model Context Protocol (MCP) to orchestrate specialized sub-agents (UI Interaction, Shell, App Research) for planning and executing complex multi-app workflows.

ENGINEERING EXPERIENCE

New York Life Insurance

New York, NY

Software Engineer Intern, TDAV Team

May – Aug 2026

- Created an AI text and voice agent for processing insurance claims for clients and 12,000 Customer Service Representatives using AWS Infrastructure with access to NYL services, knowledge, and forms.
- Constructed a company wiki/knowledge base from 13,000 documents compiled from Sharepoint, Public Site, Confluence, Jira, and the Intranet bringing business logic into one place for AI and human users.

Lykke

San Francisco, CA

Founding Software Engineer

Sep 2025 – Present

- Built core AI workflows for an AI study platform used to ingest Canvas LMS courses, assignments, announcements, notes, and uploaded documents into class-scoped knowledge bases for contextual chat and study generation.
- Engineered adaptive RAG pipelines across MongoDB, Zilliz/Milvus, S3, Gemini Files API, and SSE streaming, including source inference, citation events, token-budgeting for large course materials, and fallback paths for extracted text vs. full-file reasoning.

Scarlet Sync

New Brunswick, NJ

Founder

Jan 2025 – Present

- Architected a robust data ingestion pipeline using Python and LLMs to reverse-engineer legacy data, indexing over 2,500 classes and 500+ degree programs for real-time querying.
- Developed a production AI assistant with streaming tool-calling and grounded retrieval over schedules, degree plans, and profile context, enabling automated schedule building and personalized academic planning across Web, ChatGPT, Claude, and iOS.

Samaritan Scout

Cranford, NJ

Full Stack Engineer Intern

May 2023 – Aug 2025

- Developed an agentic scraping system leveraging OpenAI and Gemini structured outputs to autonomously extract data from over 100,000 non-profit websites, reducing manual data entry time by over 90%.
- Developed a full-stack search engine using React, TypeScript, and PostgreSQL to index 35,000+ volunteer opportunities, implementing semantic search algorithms with hallucination filters to ensure accurate matching.

AWARDS & SKILLS

Awards: 8090 AI Top Coder (5th), Rutgers Shark Tank (2nd and 3rd), ASA Fall Data Challenge (Top 3), NSF I-Corps.

Languages: Python, C++, CUDA, Java, SQL, TypeScript, R, Bash, Git

Technologies: PyTorch, CUDA, Transformers, Quantization & Pruning, HPC Clusters, Generative AI, Agents, Scikit-learn, Pandas, AWS